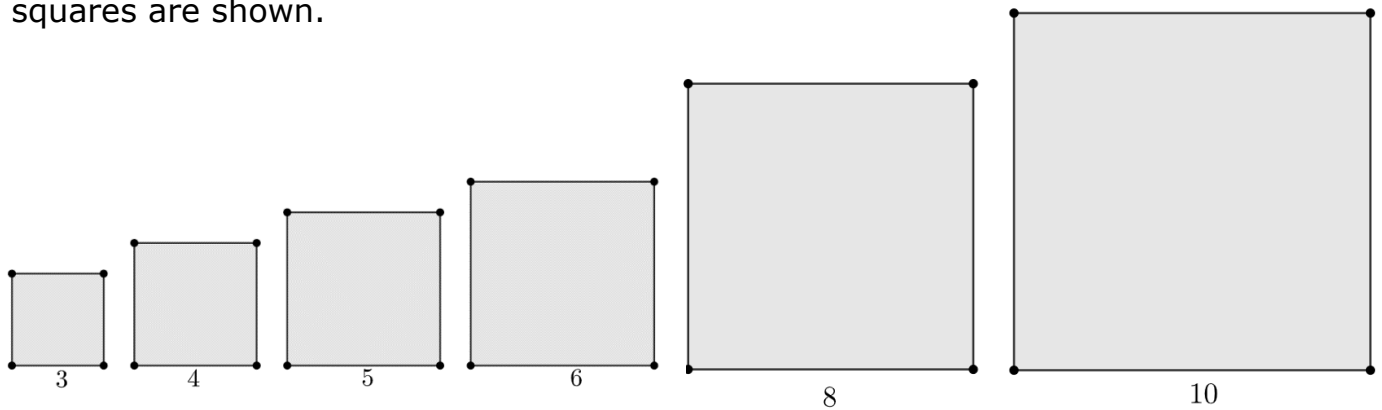


Geometry
Unit 7
Lesson 1 Practice

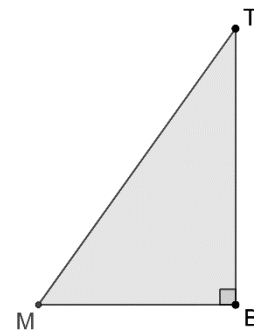
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Six squares are shown.



The side measures of right triangle NWC are: $WC = 6$, $WN = 8$, and $NC = 10$.

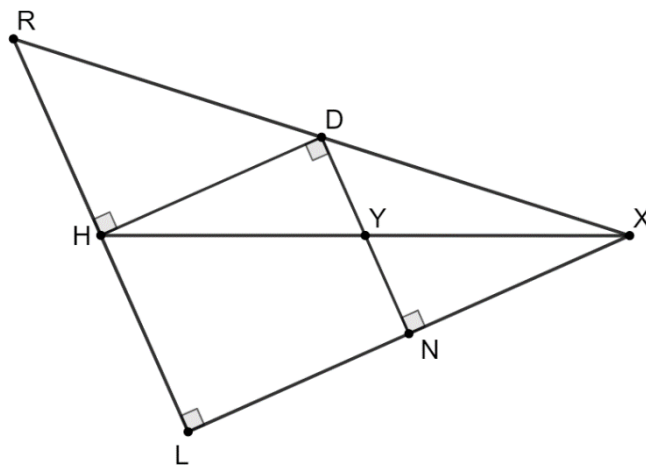
1. To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WN}$?
2. To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WC}$?
3. $\triangle MTB$ is shown where $MT = \sqrt{74}$ and $MB = 5$. Find the exact length of $\overline{BT}$.



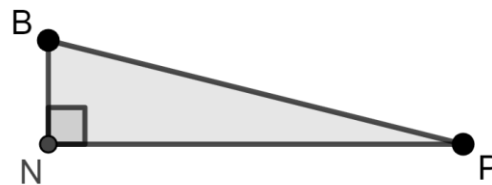
4. A right triangle has side lengths of $6\sqrt{3}$ and $\sqrt{15}$. Find the exact length of the hypotenuse.

5. Triangle RLX is shown where $LN = NX$, $RL = 8.766$, $XR = 13.184$, and $YX = 5.39$.

Determine the length of $\overline{NY}$. If necessary, round your answer to the nearest thousandth.



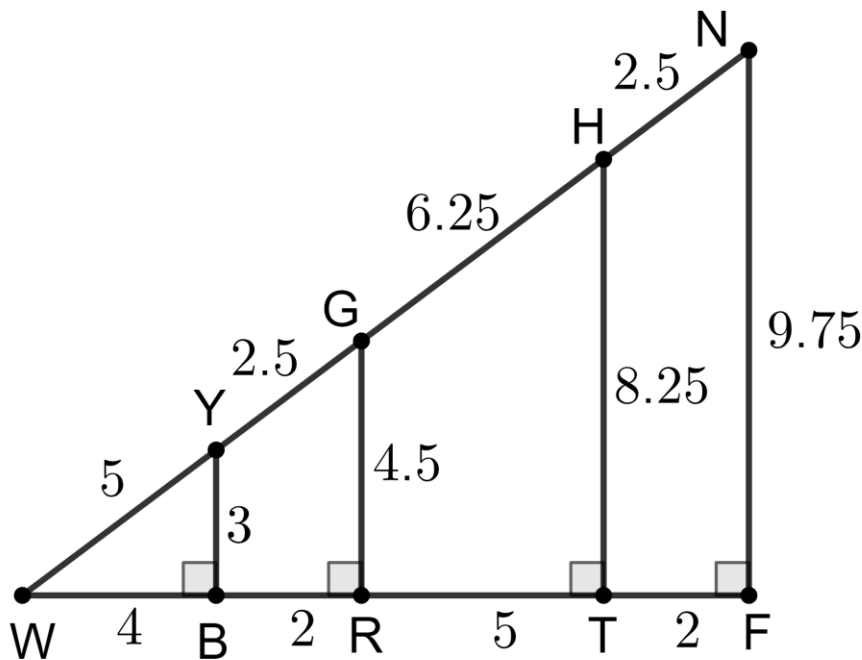
$\triangle BNP$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table.



6.
7.
8.
9.
10.

	Length of $\overline{BN}$	Length of $\overline{NP}$	Length of $\overline{BP}$
6.	48		73
7.		5	$\sqrt{42}$
8.	7	$\sqrt{95}$	
9.	33		65
10.		$\sqrt{2}$	$\sqrt{29}$

The diagram shows measurements for $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$.



Complete the table using $\angle W$ as the reference angle.

	Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
1.	$\triangle YBW$			
2.	$\triangle GRW$			
3.	$\triangle HTW$			
4.	$\triangle NFW$			

For $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$ find the decimal equivalent for each ratio.

	Ratio	$\triangle YBW$	$\triangle GRW$	$\triangle HTW$	$\triangle NFW$
5.	$\frac{\text{opposite}}{\text{hypotenuse}}$				
6.	$\frac{\text{adjacent}}{\text{hypotenuse}}$				
7.	$\frac{\text{opposite}}{\text{adjacent}}$				

Complete the statements.

8. The decimal equivalents of the

- ☐ $\frac{\text{opposite}}{\text{adjacent}}$
- ☐ $\frac{\text{adjacent}}{\text{hypotenuse}}$
- ☐ $\frac{\text{opposite}}{\text{hypotenuse}}$
- ☐ three

ratios for triangles YBW ,

GRW , HTW , and NFW are equal.

9. Because triangles YBW , GRW , HTW , and NFW are

- ☐ congruent
- ☐ similar

triangles,

their side lengths are

- ☐ congruent.
- ☐ proportional.

10. Since triangles YBW , GRW , HTW , and NFW share

- ☐ $\angle W$
- ☐ side $\overline{FN}$
- ☐ sides $\overline{FN}$ and $\overline{RN}$

and

are

- ☐ acute
- ☐ obtuse
- ☐ right

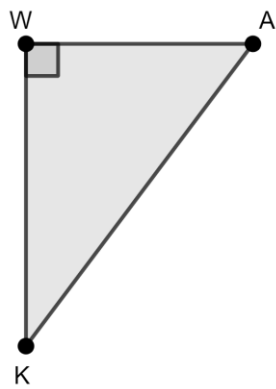
triangles, it is possible to define trigonometric ratios in any right

triangle.

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Triangle WKA is shown where $m\angle W = 90^\circ$.

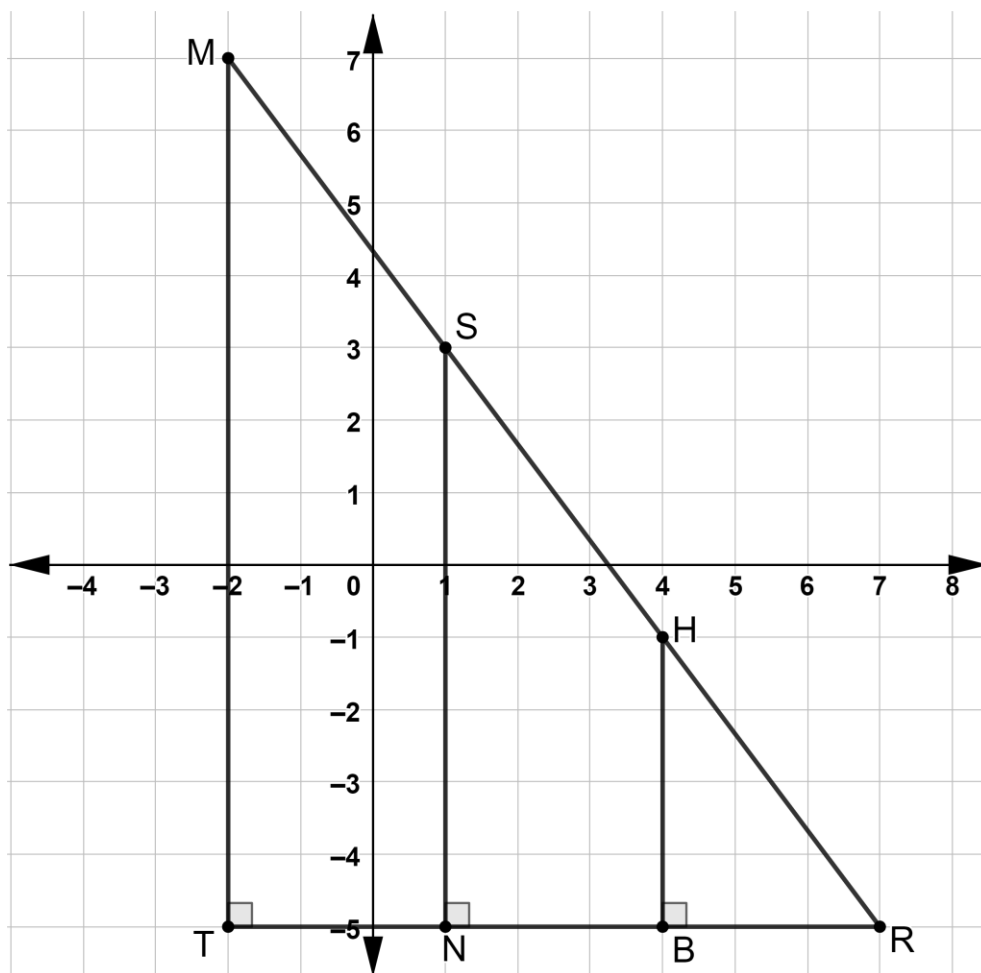


Complete the table using the given information.

	$\overline{WA}$	$\overline{WK}$	$\overline{KA}$	Reference Angle	Cosine	Sine	Tangent
1.	36		85	$\angle A$			
2.	20	21		$\angle K$			
3.	20	21		$\angle A$			
4.		$\sqrt{51}$	$\sqrt{82}$	$\angle K$			

	$\overline{WA}$	$\overline{WK}$	$\overline{KA}$	Reference Angle	Cosine	Sine	Tangent
5.		$\sqrt{51}$	$\sqrt{82}$	$\angle A$			
6.	$4\sqrt{17}$	20		$\angle K$			
7.	$6\sqrt{2}$		11	$\angle A$			
8.				$\angle K$	$\frac{84}{85}$	$\frac{13}{85}$	
9.				$\angle A$			$\frac{11}{8}$
10.				$\angle K$			$\frac{9}{40}$

Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$M(-2, 7)$

$S(1, 3)$

$H(4, -1)$

$R(7, -5)$

$T(-2, -5)$

$N(1, -5)$

$B(4, -5)$

1. Explain why these triangles are similar.

Complete the table.

	Triangle	Side	Length
2.	HRB	$\overline{BH}$	
		$\overline{BR}$	
		$\overline{RH}$	
3.	SRN	$\overline{NS}$	
		$\overline{RN}$	
		$\overline{SR}$	
4.	MTR	$\overline{MT}$	
		$\overline{RT}$	
		$\overline{MR}$	

Using $\angle R$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	HRB			
6.	SRN			
7.	MTR			

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$\cos(53^\circ)$	
9.	$\sin(53^\circ)$	
10.	$\tan(53^\circ)$	

Geometry
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$\triangle PGL$ is a right triangle where $m\angle LPG = 90^\circ$, $LP = 8$, and $LG = 17$.

1. Determine $m\angle LGP$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle BLC$ is a right triangle where $m\angle BCL = 90^\circ$, $BC = 33$, and $LC = 56$.

2. Determine $m\angle CBL$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle MDZ$ is a right triangle where $m\angle DZM = 90^\circ$, $MD = 41$, and $DZ = 9$.

3. Determine the length of $\overline{ZM}$. If necessary, round your answer to the nearest thousandth.
4. Determine $m\angle MDZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle TKB$ is a right triangle where $m\angle KTB = 90^\circ$, $KT = 92$, and $KB = 115$.

5. Determine the length of $\overline{TB}$. If necessary, round your answer to the nearest thousandth.

6. Determine $m\angle TKB$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle SPC$ is a right triangle where $m\angle CSP = 90^\circ$, $m\angle CPS = 27^\circ$, and $CS = 14$.

7. Determine the length of $\overline{CP}$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

8. Explain why the tangent ratio cannot be used to determine the length of $\overline{CP}$.

$\triangle NLT$ is a right triangle where $m\angle NTL = 90^\circ$, $m\angle TNL = 33^\circ$, and $NL = 157$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.

10. Determine the length of $\overline{NT}$. Show your work. If necessary, round your answer to the nearest thousandth.

Geometry
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Name _____
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$\triangle BXC$ is a right triangle where $m\angle XCB = 90^\circ$, $m\angle CXB = 60^\circ$, and $XC = 5$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
1.	$\overline{CB}$	
2.	$\overline{XB}$	

$\triangle YFW$ is a right triangle where $m\angle YFW = 90^\circ$, $m\angle FYW = 45^\circ$, and $FY = 3$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
3.	$\overline{WF}$	
4.	$\overline{YW}$	

$\triangle DZH$ is a right triangle where $m\angle ZDH = 90^\circ$, $m\angle DHZ = 30^\circ$, and $ZH = 4$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
5.	$\overline{ZD}$	
6.	$\overline{DH}$	

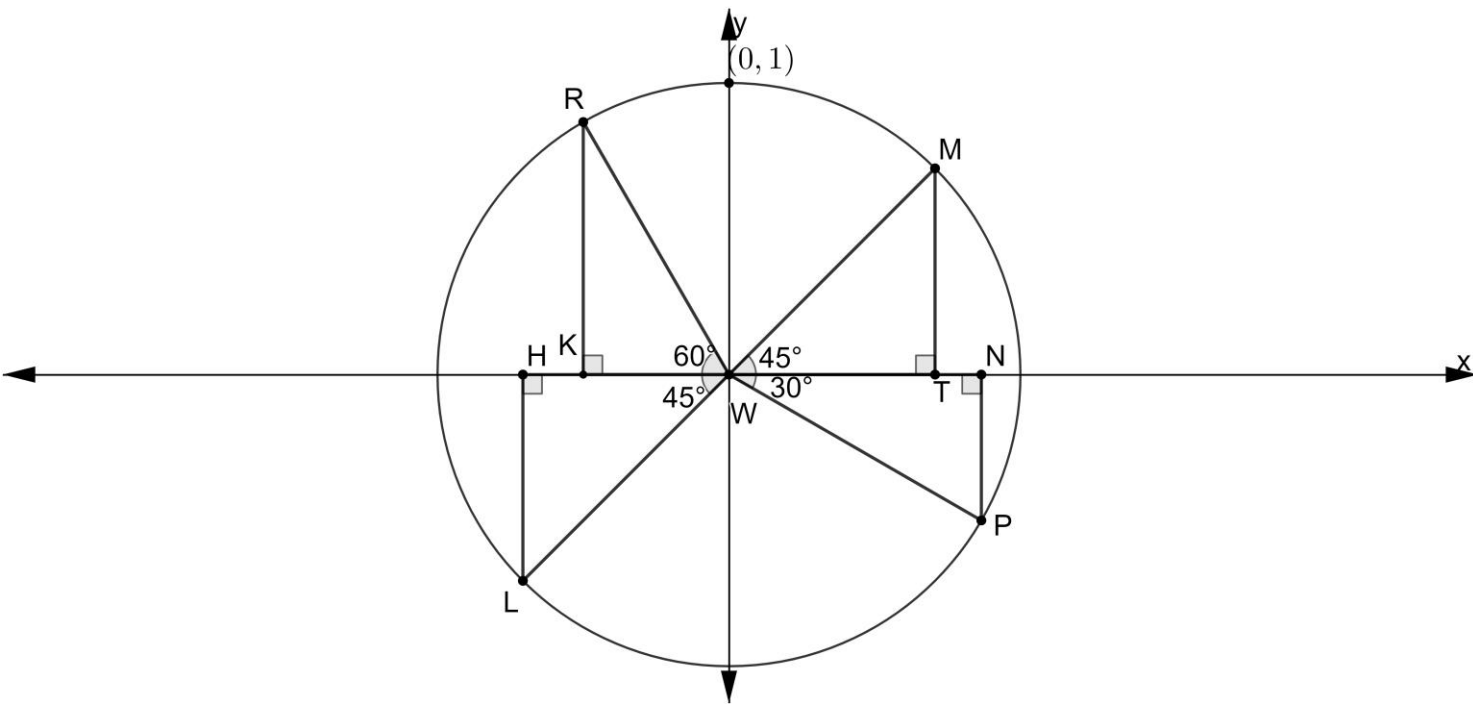
$\triangle GVU$ is a right triangle where $m\angle GVU = 90^\circ$, $m\angle UGV = 45^\circ$, and $VG = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{VU}$	
8.	$\overline{GU}$	

$\triangle JTS$ is a right triangle where $m\angle TSJ = 90^\circ$, $m\angle JTS = 60^\circ$, and $TJ = 12$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{TS}$	
10.	$\overline{SJ}$	

Circle W and right triangles WMT , WPN , RWK , and LWH are shown on the coordinate plane.



1. What is the length of the radius of circle W ?

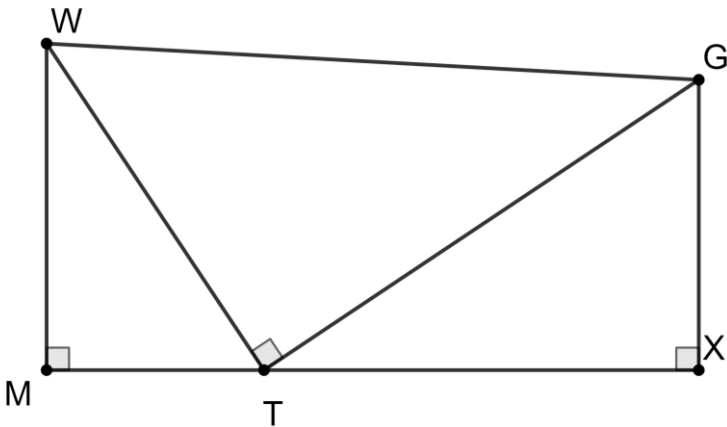
Complete the table.

	Segment	Length
2.	$\overline{WM}$	
3.	$\overline{PW}$	

Complete the table.

	Point	Coordinates
4.	L	
5.	P	
6.	K	
7.	R	
8.	N	
9.	M	
10.	H	

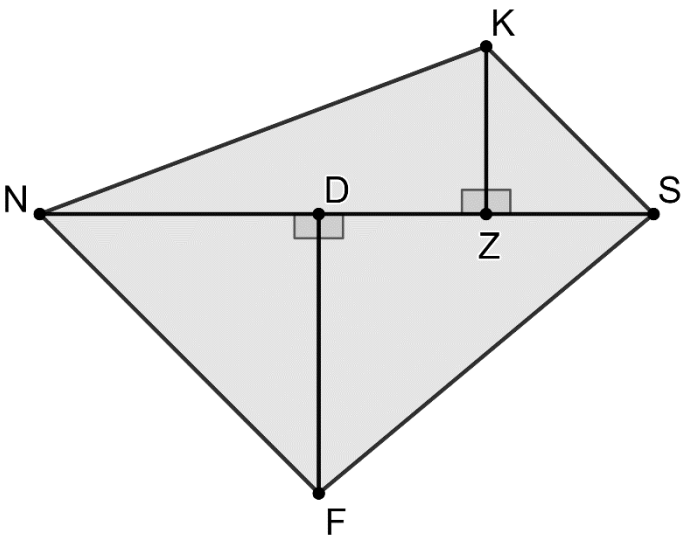
Right triangles WMT , GTW , and TXG are shown where $TX = 8$, $m\angle XTG = 34^\circ$, and $m\angle TGW = 37^\circ$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. If necessary, round to the nearest thousandth.

	Side	Work	Side Measure
1.	$\overline{XG}$		
2.	$\overline{TG}$		
3.	$\overline{WT}$		
4.	$\overline{WM}$		
5.	$\overline{MT}$		

Quadrilateral $NKSF$ is composed of right triangles NDF , FDS , SZK , and NZK . $ZK = 3$, $DF = 5$, $m\angle DNF = 45^\circ$, $m\angle SKZ = 45^\circ$, $m\angle FSD = 40^\circ$, and $m\angle ZNK = 21^\circ$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure. If necessary, round to the nearest thousandth.

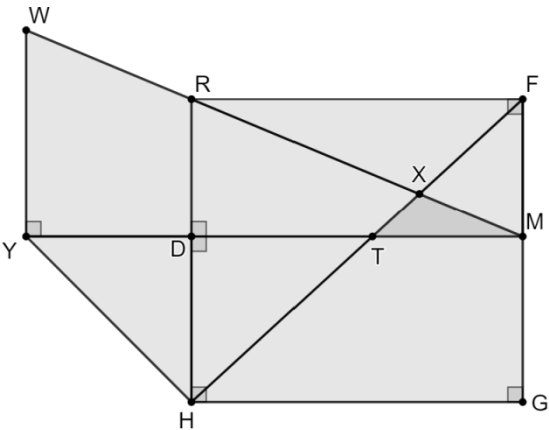
	Side	Work	Side Measure
6.	$\overline{KS}$		
7.	$\overline{FS}$		
8.	$\overline{NF}$		
9.	$\overline{NK}$		

10. What is the perimeter, to the nearest thousandth, of quadrilateral $NKSF$?

Alafia River State Park has a set of mountain bike trails. Mario draws out an area of the trails that he plans to ride. His diagram is shown.

The table shows known measurements of the trails.

$m\angle DYH = 45^\circ$	$YW = 5$ kilometers
$m\angle XMT = 23^\circ$	$DM = 8$ kilometers
$HD = 4$ kilometers	$\triangle FMR \cong \triangle DRM$
$m\angle GHF = 43^\circ$	

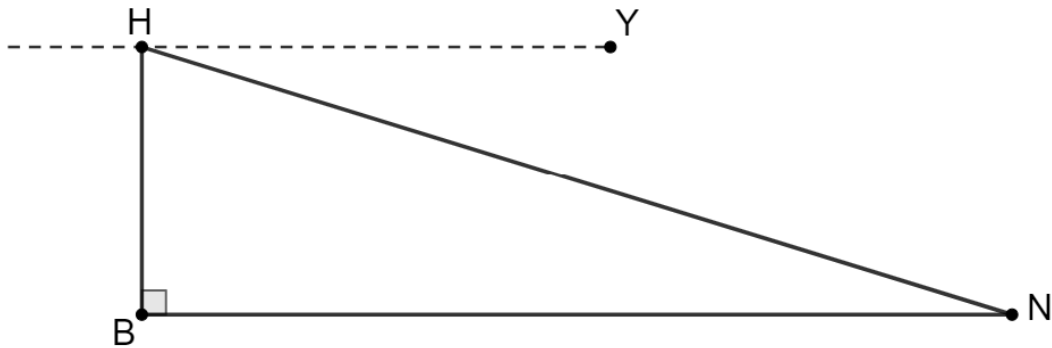


Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. Round to the nearest tenth.

	Side	Work	Side Measure
1.	$\overline{HF}$		
2.	$\overline{HY}$		
3.	$\overline{RD}$		
4.	$\overline{RM}$		

5. Mario starts his ride at point W and rides along the perimeter of the park. What is the total distance, in kilometers, Mario rides? Round to the nearest tenth of a kilometer.
6. What is the total distance, in kilometers, if Mario rides from W to Y to H to T to X to F to R ending at W . Round to the nearest tenth of a kilometer.

Gabby is flying a drone over Lake Jackson. The drone is represented by point H and is 265 feet above Lake Jackson which is represented by point B . The angle of depression, $\angle YHN$, is the angle that Gabby will fly the drone for it to land on the fishing pier, represented by point N . The measure of $\angle YHN$ is 17° .



7. Explain why $\angle YHN$ is congruent to $\angle BNH$.

8. Determine the horizontal distance, BN , the drone is from the fishing pier. Round your answer to the nearest tenth.

9. Determine the distance, HN , the drone must travel to reach the fishing pier. Round your answer to the nearest tenth.

10. Zac is also flying a drone over Lake Jackson. His drone is 11,000 feet across Lake Jackson horizontally from the fishing pier. If the height of his drone is 400 feet, what is the angle of depression Zac should fly his drone so it lands on the fishing pier? Round your answer to the nearest tenth of a degree.

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In $\triangle B K X$, $m\angle K B X = 19^\circ$, $m\angle B X K = 41^\circ$, and $X K = 32$. Determine the measurement of each part of $\triangle B K X$. Show your work below the table. If necessary, round to the nearest tenth.

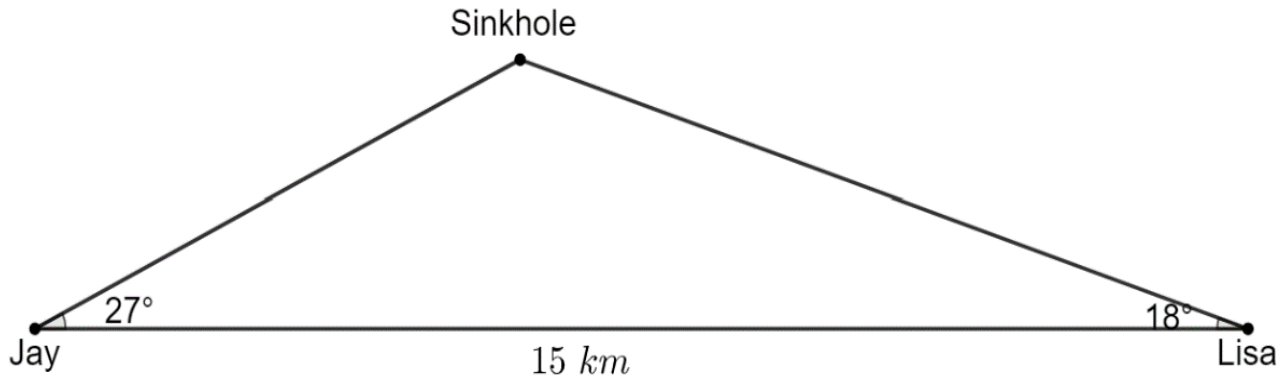
	Part	Measurement
1.	$m\angle X K B$	
2.	$\overline{X B}$	
3.	$\overline{K B}$	

In $\triangle Z F S$, $m\angle S Z F = 47^\circ$, $Z F = 28.3$ and $S F = 51.4$. Determine the measurement of each part of $\triangle Z F S$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$m\angle F S Z$	
5.	$m\angle S F Z$	
6.	$\overline{S Z}$	

Jay and Lisa are both hiking in the Leon Sinks Geological Area. They are currently 15 kilometers (km) apart. They plan to meet up at one of the sinkholes in the park.

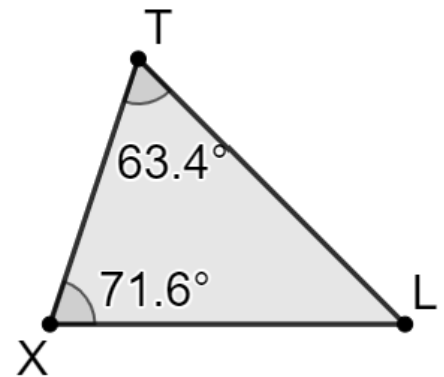
7. How far, to the nearest hundredth of a kilometer, does Jay hike?



8. How far, to the nearest hundredth of a kilometer, does Lisa hike?

Shanice runs in a competitive obstacle course. The obstacle course and some of its measurements are shown. The distance between point T and point X is 3.2 miles.

9. Write an equation that can be used to determine the distance, in kilometers, between point T and point L .



10. Determine the perimeter, in miles, of the obstacle course.

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In $\triangle GHS$, $m\angle HGS = 22^\circ$, $HG = 27$, and $SG = 14.9$. Determine the measurement of each part of $\triangle GHS$. Show your work below the table. If necessary, round to the nearest tenth.

- 1.
- 2.
- 3.

Part	Measurement
$m\angle SHG$	
$m\angle HSG$	
$\overline{HS}$	

In $\triangle PXN$, $PX = 17$, $PN = 28$, and $XN = 30$. Determine the measurement of each part of $\triangle PXN$. Show your work below the table. If necessary, round to the nearest tenth.

- 4.
- 5.
- 6.

Part	Measurement
$m\angle PXN$	
$m\angle PNX$	
$m\angle XPN$	

Austyn is checking a water trough in her pasture before walking to the barn. She enters the pasture through a gate near her house. The distance from the gate, point G , to the water trough, point W , is 52 yards. The distance from the gate to the barn, point N , is 37.5 yards. The distance from the water trough to the barn is 31 yards. Determine each angle measure.

7.
8.
9.

Angle	Measurement
$\angle NGW$	
$\angle GWN$	
$\angle WNG$	

10. In $\triangle DTF$, $DF = 10$, $TD = 42$, and $TF = 46$. Complete the statement about $\triangle DTF$.

$\triangle DTF$ is a

- ☐ right
 ☐ acute
 ☐ obtuse
 ☐ right

triangle because

- ☐ one of the angles is
 ☐ all of the angles are

- ☐ equal to 90° .
 ☐ less than 90° .
 ☐ greater than 90° .

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Determine the area of each triangle using the given information. Round to the nearest tenth.

	Triangle	Given Information			Area
1.	$\triangle BHZ$	$HB = 7$	$ZB = 12$	$m\angle HBZ = 64^\circ$	
2.	$\triangle YCJ$	$CY = 100$	$JY = 63$	$m\angle JYC = 80^\circ$	
3.	$\triangle KDN$	$DN = 7.2$	$NK = 6.3$	$m\angle DNK = 95^\circ$	
4.	$\triangle FSG$	$SF = 3\sqrt{5}$	$GS = 2\sqrt{7}$	$m\angle FSG = 36^\circ$	

	Triangle	Given Information			Area
5.	$\triangle WQL$	$QW = 8.4$	$LW = 9.9$	$m\angle QWL = 68^\circ$	
6.	$\triangle BRS$	$SR = \frac{78}{5}$	$RB = \frac{119}{8}$	$m\angle SRB = 42^\circ$	
7.	$\triangle TXD$	$TX = 2.6$	$TD = 5.1$	$m\angle XTD = 16^\circ$	
8.	$\triangle YNK$	$NY = \sqrt{11}$	$NK = \sqrt{11}$	$m\angle KNY = 72^\circ$	
9.	$\triangle LGR$	$GL = 521$	$LR = 94$	$m\angle RLG = 179^\circ$	
10.	$\triangle ABC$	$BC = 5\sqrt{3}$	$AB = 4\sqrt{7}$	$m\angle ABC = 89^\circ$	